

Draft - expert review pending

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Cognitive Processes in Learning - assessment and review

Unit summary

Attention is the gateway: a narrow spotlight, captured by novelty, contrast, meaning and emotion rather than commanded, costly to divide, and depletable. A lesson competes for the beam with structure or on the distractors' own terms, and the beam's design is a teacher's decision.

Perception is active interpretation: what a pupil sees depends on what they already know, ambiguous materials are read through expectation, and expert perception is learned through named encounters.

Memory is a chain of systems with jobs: sensory memory selects, working memory holds a handful of items for seconds and thinks with them, long-term memory keeps what encoding has connected. Storage strengthens with spacing, retrieval with practice, and forgetting has three faces - decay, interference, retrieval failure - each with its own remedy. The strategies the evidence supports are the ones that make memory do its own work: spaced retrieval, elaboration, teaching it to someone.

Thinking runs on concepts and named strategies - algorithm, heuristic, means-ends, working backwards, analogy - against obstacles that belong to everyone: functional fixedness, set effect, overload. Reasoning runs from cases to rules (inductive) and rules to cases (deductive), and each fails differently.

Summative assessment

Multiple-choice questions (MCQs)

1. Attention is best described as: (*Remember*) a) the storage of information for later use b) the mind's selection of what gets through to further processing c) the interpretation of what the senses deliver d) a fixed quantity a pupil has or lacks
2. "Copy this while I explain" tends to produce notes without understanding because: (*Understand*) a) copying is too easy to be engaging b) the two tasks are competing for one narrow workbench c) pupils cannot write and hear at the same age d) the explanation is always the weaker task
3. Sensation differs from perception in that sensation is: (*Understand*) a) interpretation and perception is detection b) detection by a receptor, while perception organises and interprets what was detected c) conscious and perception is automatic d) longer-lasting than perception
4. The practical capacity of working memory is closest to: (*Remember*) a) two items b) four items c) twelve items d) unlimited
5. The chapter re-read four times in one night is gone by December mainly because: (*Apply*) a) decay: the trace was never connected to anything b) it was never encoded with connections, and no retrieval route was practised c) interference from the November chapter erased it d) long-term memory rejected it as unimportant
6. Which study strategy has the strongest evidence for durable recall? (*Understand*) a) re-reading b) highlighting c) spaced retrieval practice d) listening once attentively
7. Re-reading feels effective because it creates: (*Understand*) a) recall b) recognition c) elaboration d) interference
8. Ayesha draws the rickshaw problem as two bars. The strategy is: (*Apply*) a) algorithm b) analogy c) means-ends analysis made visible d) deduction

9. "Every rickshaw fare I have checked ended in a zero" is: (*Understand*) a) deductive reasoning b) inductive reasoning c) a heuristic failure d) retrieval failure
10. Bilal's "use the numbers" habit is best described as: (*Apply*) a) functional fixedness b) a set effect: a habit that solved yesterday's problems c) an algorithm d) divided attention

Restricted-response questions (RRQs)

1. Explain attention's three properties, with a classroom example of each. (*Understand*)
2. In two sentences, explain why the teacher's "it's obvious, look" fails, using perception. (*Understand*)
3. Describe the three memory systems in order, explaining the job of each. (*Understand*)
4. A lesson gives a four-step instruction and then asks a question. Explain why the question is often not heard, using working memory. (*Apply*)
5. Explain, in one line each, what encoding, storage and retrieval do for a memory. (*Understand*)
6. Name the three faces of forgetting and the remedy each answers to. (*Understand*)
7. Explain recognition versus recall, and why an examination asks for the second. (*Understand*)
8. Explain any three problem-solving strategies by giving an instance of each. (*Understand*)
9. Distinguish inductive from deductive reasoning with one classroom example of each. (*Understand*)
10. In three lines, design a class activity that works with attention and memory, naming the rule each line follows. (*Apply*)

Extended-response questions (ERQs)

1. **The timetable move.** Miss Hina moved the hardest listening to the first period and the copying to the noisy hour, and reading marks rose. Using attention's properties, analyse why it worked, what it cost, and name one further change using a different property. (*Analyze*)

2. **The October chapter.** A trainee learns Pakistan Studies by re-reading on Monday night, tests well, and has lost it by December. Using the memory model, analyse what was never done, and design the two-week revision plan the evidence supports, naming what each element asks of the memory systems. (*Analyze*)
3. **Bilal's 64.** Analyse Bilal's answer as a thinking problem rather than an ability verdict: the obstacle, the missing strategies, and what his habit says about the problems he has been set. Design the next lesson's first ten minutes and justify each element from the unit. (*Analyze*)
4. **The strategy audit.** Audit your own three study strategies against the evidence of this unit: what each asks of memory, what the evidence says, and the single change you would make. Honesty about your own habits is the criterion. (*Evaluate*)
5. **The whole unit in one lesson.** Design one lesson - any class, any subject - that works with the whole unit: the beam aimed and one demand at a time, the workbench load fitted and offloaded, a strategy named aloud, and retrieval spaced into the week. Justify every design choice by the rule it follows, and name the cost. (*Create*)

Answers and marking guidance

MCQ answer key

1. **b** - attention selects what reaches further processing; it is the gateway.
2. **b** - two attention-heavy tasks on one workbench: the copying wins, the explaining gets the remainder.
3. **b** - detection versus organised interpretation.
4. **b** - recent estimates put practical capacity near four items.
5. **b** - no connections at encoding, no practised route out: it was never stored in a reachable form.
6. **c** - retrieval practice, spaced, is the best-evidenced of the set.
7. **b** - re-reading creates recognition, which feels like knowing.

8. **c** - comparing where you are with where you must be, step by step: means-ends.
9. **b** - from cases to a rule: induction.
10. **b** - a set effect: the habit generalises from yesterday's same-shaped problems.

RRQ model answers and mark schemes

Award marks for the substance below, not for matching wording.

1. Captured not commanded; costly to divide; depletable (1 for all three named); any apt examples (1).
2. The teacher's perception is expert: they have seen the diagram a hundred times, so its difficulty is invisible to them (1); the class reads it through expectation and missing concepts, so it is a puzzle - the remedy is naming the parts (1).
3. Sensory memory selects impressions for a second or two; working memory holds and works with a handful of items for seconds; long-term memory keeps what encoding connected (1 for all three; 1 for any two jobs accurate).
4. The four steps already occupy the workbench; the question arrives with no space to run in (2).
5. Encoding: input with connections; storage: retention, strengthened by spacing; retrieval: getting it back, strengthened by practice (1 for all three).
6. Decay answers spacing; interference answers distinctiveness; retrieval failure answers practice (1 for the three faces, 1 for matching remedies).
7. Recognition is "I have seen this"; recall is "I can produce this" (1); an examination asks for production, so only recall counts (1).
8. Any three of algorithm, heuristic, means-ends, working backwards, analogy, with instances (2: 1 for three named, 1 for instances).
9. Inductive: cases to rule, e.g. "this teacher always marks late"; deductive: rule to case, e.g. "if the test is Friday and today is Thursday..." (2).
10. Any three-line activity in which each line names its rule: e.g. question on the board first (aim the beam); givens written down (offload the workbench); strategy named aloud (concept formation) (2: 1 for the activity, 1 for the rules named).

ERQ rubrics

Each ERQ is marked out of 20 using the analytic scheme below.

ERQ 1 - the timetable move

Criterion	Marks
The properties at work are named precisely, not "noise is bad"	5
The cost of the move and the second change are both concrete (<i>Analyze</i>)	7
The second change uses a genuinely different property	4
Plain, accurate, organised writing within the word count	4

ERQ 2 - the October chapter

Criterion	Marks
The analysis names what was never done (connections, retrieval routes)	5
The plan's elements map to spaced retrieval, elaboration and teaching-it (<i>Analyze</i>)	7
Each element is justified by what it asks of the systems	4
Plain, accurate, organised writing within the word count	4

ERQ 3 - Bilal's 64

Criterion	Marks
The obstacle and missing strategies are named, not "he is weak"	5

Criterion	Marks
The ten-minute design is justified element by element (<i>Analyze</i>)	7
The habit is explained by the problems he has been set	4
Plain, accurate, organised writing within the word count	4

ERQ 4 - the strategy audit

Criterion	Marks
Each of the three strategies is placed accurately on the evidence	6
The chosen change is the one the evidence most supports, and said as an action (<i>Analyze</i>)	6
Honesty: the audit is of real habits, not ideal ones	4
Plain, accurate, organised writing within the word count	4

ERQ 5 - the whole unit in one lesson

Criterion	Marks
All four design rules appear: beam, load, named strategy, spaced retrieval	6
Every choice is justified by its rule (<i>Analyze</i>)	6
The lesson is realistic for a real class, and its cost is named	4
Plain, accurate, organised writing within the word count	4

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